

JKBOSE 2026

CLASS 11TH

Morphology of Flowering Plants

LECTURE-08

BOTANY

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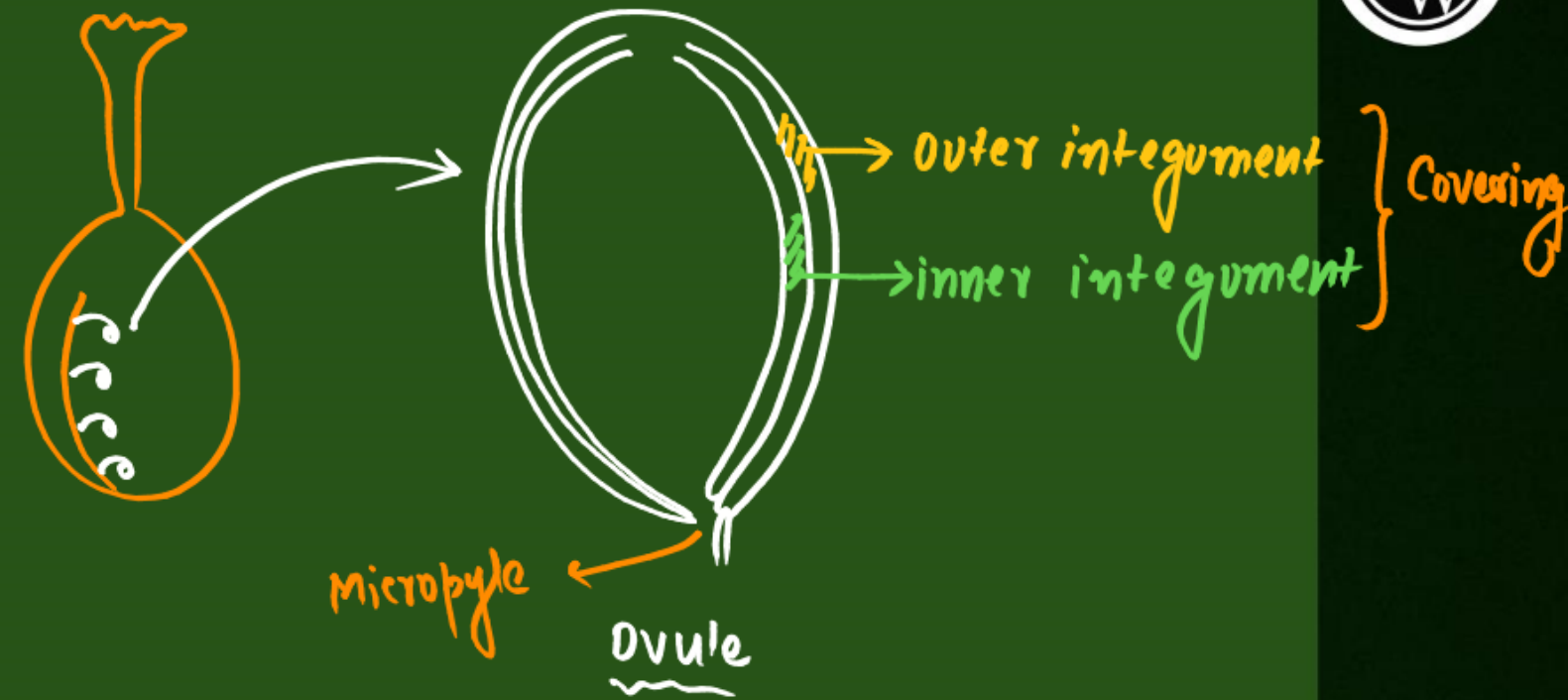
Topics To Covered

1 SEED

2 FRUIT

SEED

- ↳ Seed is formed from ovule after fertilisation.
- Seed is fertilised ovule.
- In angiosperms (flowering plants), Seeds are of two types:
 - a) Dicot Seed.
 - b) Monocot Seed.



DICOT SEED

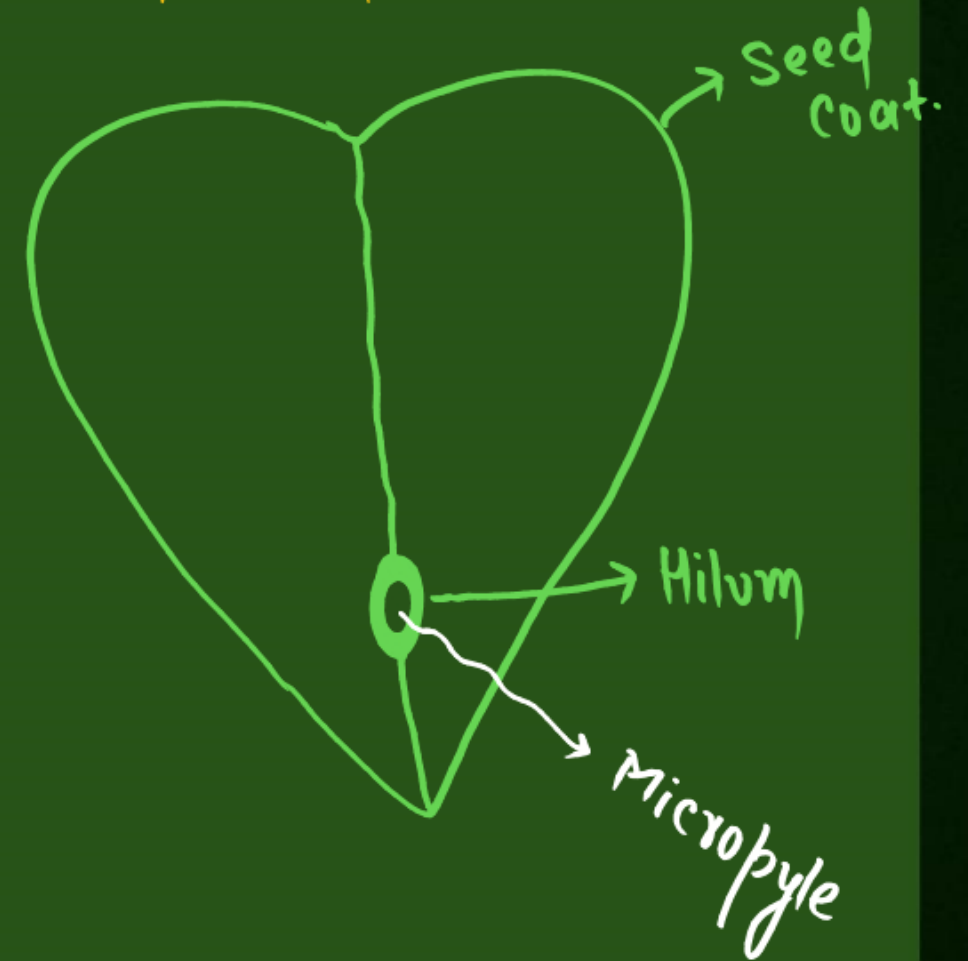
↳ It consists of:

1. Seed Coat → It is outermost covering of seed.
→ It provides protection to embryo (not inside seed coat).

↳ It develops from integuments of ovule.

↳ It is two layered →

1. Testa: Outer layer.
→ formed from outer integument of ovule.
2. Tegmen: Inner layer
→ formed from inner integument of ovule.



Hilum

It is a Scar on the Seed coat through which Seed remains attached with fruit.

Micropyle

It is a small opening above the hilum through which H_2O and O_2 transport occurs.

b) Embryo

→ It is present inside seed coat.

→ It forms new plant after germination.

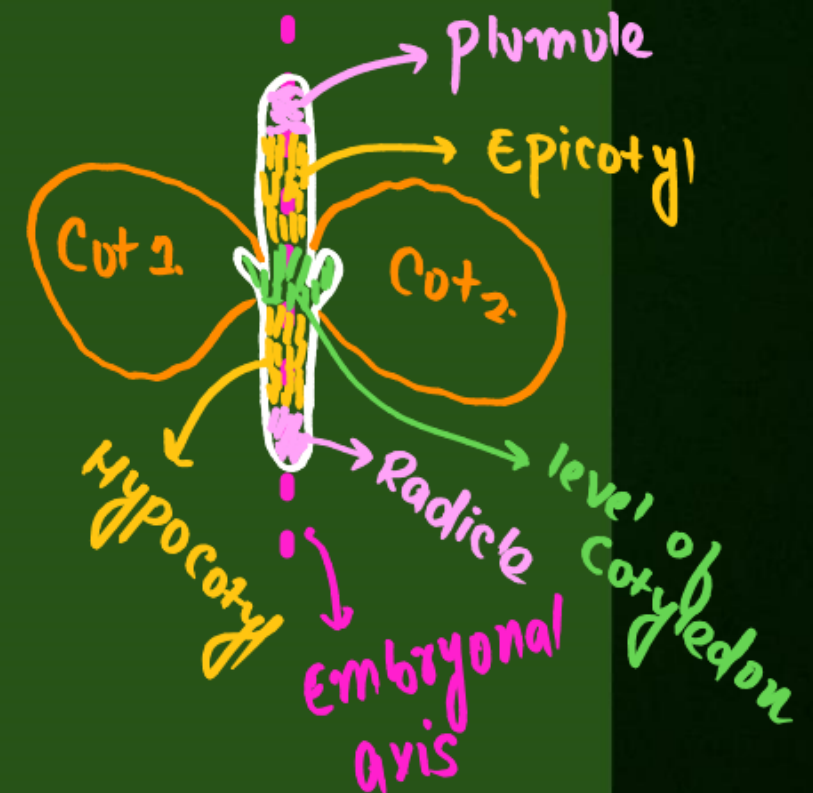
→ It consists of:

1) Embryonal axis (AKA Tigellum). → It possess two Cotyledons which possess reserve food material.

2) Two Ends of Embryonal axis are kld as Plumule and Radicle.

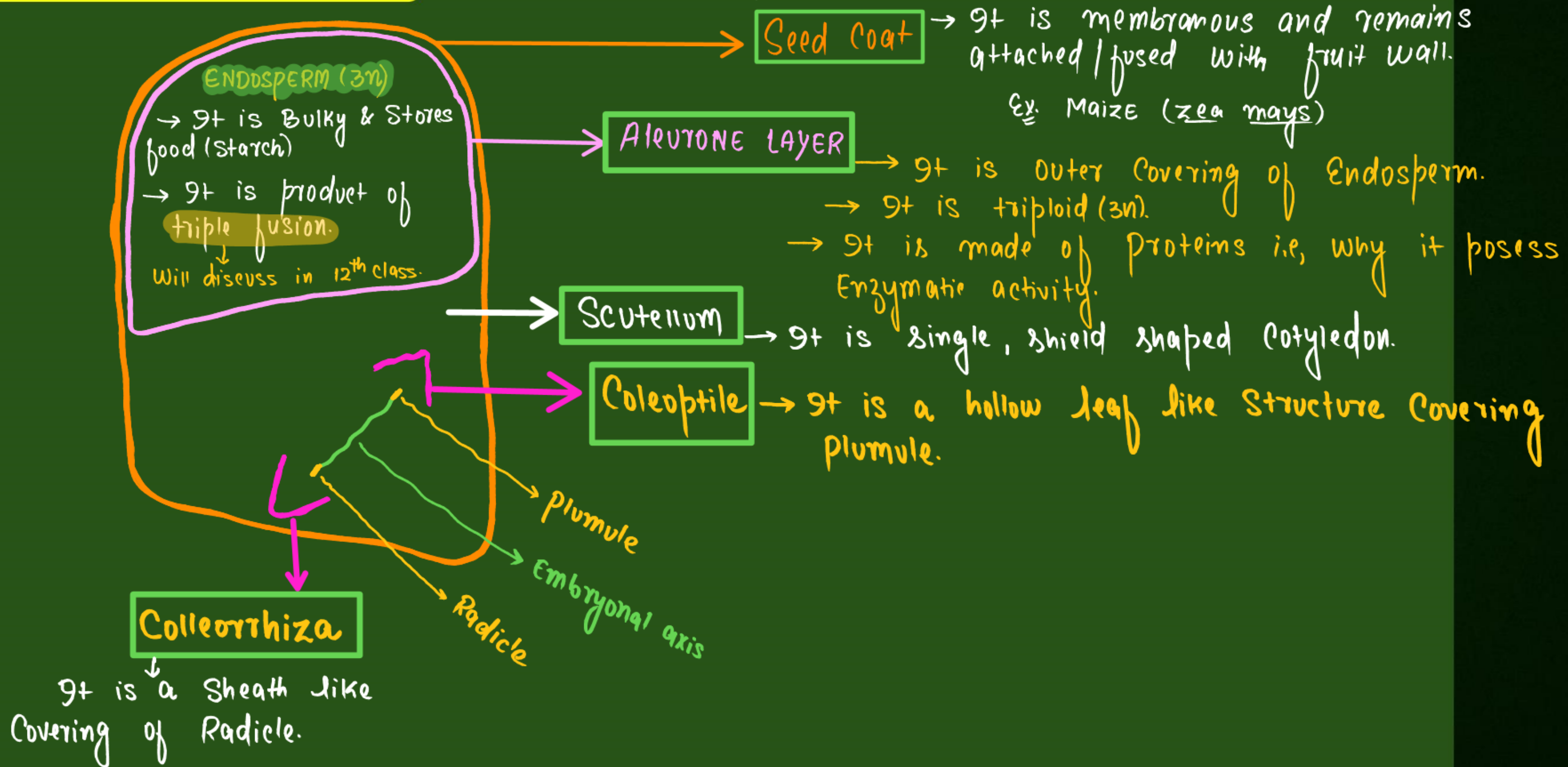
→ Plumule — forms shoot of plant.

→ Radicle — forms Root of plant.



MONOCOT SEED

Ex. Maize (zea mays)



SEED

TWO TYPES

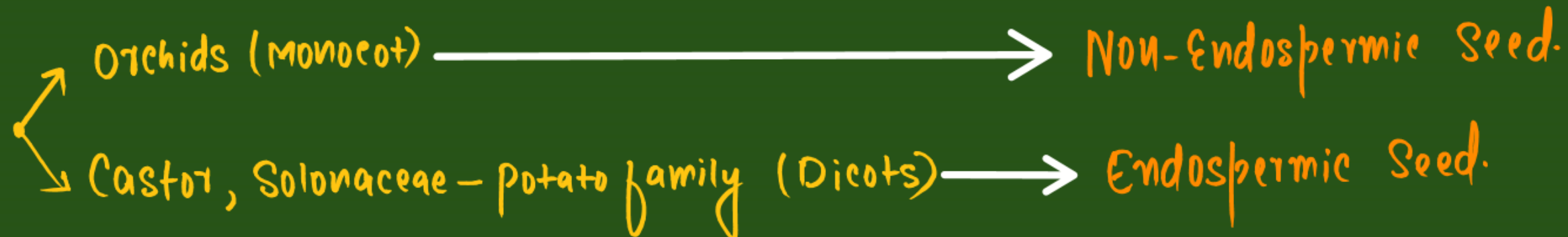
ENDOSPERMIC SEED OR ALBUMINOUS SEED

- Endosperm remains persistent even after embryo development.
- Thus in mature seed, endosperm remains present.
- It is mostly present in monocots.
- Ex: Rice, maize, Coconut, Barley.

NON-ENDOSPERMIC SEED OR EX-ALBUMINOUS SEED

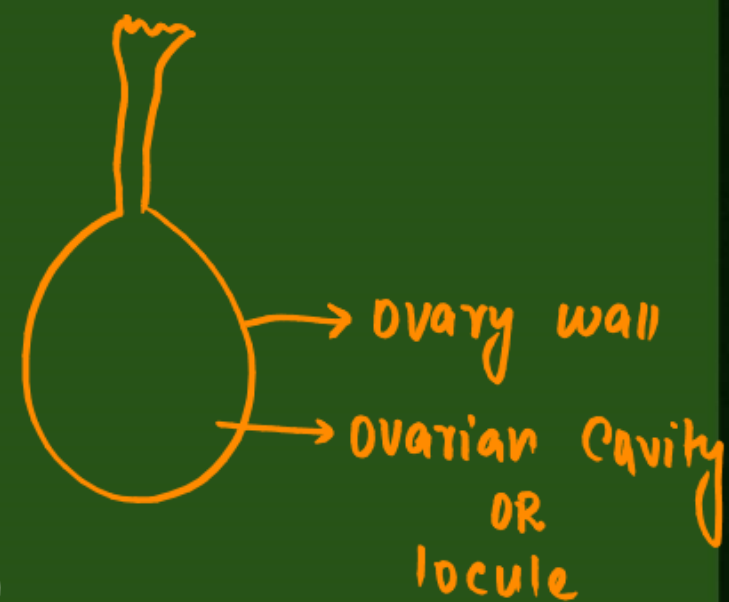
- Endosperm gets fully consumed during embryo development.
- Thus in mature seeds, endosperm remains absent.
- It is mostly present in Dicots.
- Ex: pea, Bean, Gram, Groundnut.

NOTE



FRUIT

- ↳ It is formed from ovary after fertilisation.
- ↳ Fruit is mature / ripened ovary.
- ↳ Fruit wall (AKA **pericarp**) — develops from ovary wall.
- ↳ If pericarp is thick and fleshy, it is differentiated into 03 layers
 - a. **epicarp** — **outermost** layer.
 - b. **mesocarp** — **middle** layer.
 - c. **endocarp** — **inner** layer.



Rest - we will discuss it tomorrow???

QUESTIONS

Scutellum of Maize is:

(2010 Pre)

- A** Cotyledon
- B** Endosperm
- C** Tegmen
- D** Testa

QUESTIONS

The aleurone layer in maize grain is specially rich in:

(2003)

- A** Proteins
- B** Starch
- C** Lipids
- D** Auxins

H/W

QUESTIONS

Identify the part of the seed from the given figure which is destined to form root when the seed germinates. (2024)

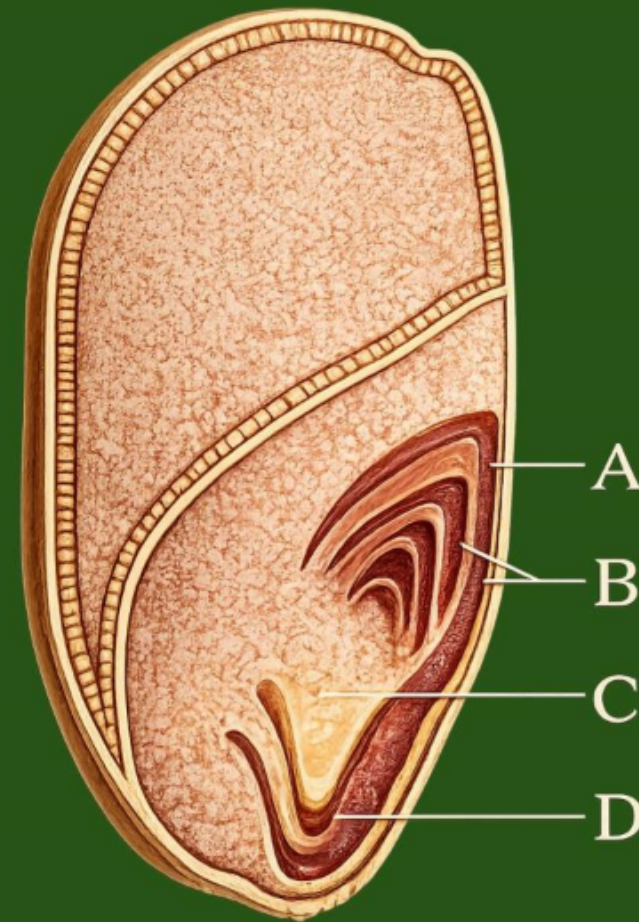
→ Radicle

A C

B A

C D

D B



Thank
You